

FUNCTIONAL REQUIREMENTS DOCUMENT

Online Food Ordering System

Web Application — Customer, Restaurant & Delivery Portal

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1. Introduction & Project Overview

1.1 Purpose of This Document

This Functional Requirements Document (FRD) defines the complete set of functional behaviors expected from the Online Food Ordering System (OFOS). It serves as the primary reference for designers, developers, testers, and business stakeholders throughout the development lifecycle. Every feature, user interaction, business rule, and integration point is described in sufficient detail for unambiguous implementation.

1.2 Product Vision

The Online Food Ordering System is a multi-sided web platform that connects customers hungry for great food with local restaurants and delivery partners. The platform enables customers to discover restaurants, explore menus, customize orders, pay securely, and track deliveries in real time — all from a single, seamless interface. Restaurant owners gain a self-service portal to manage menus, promotions, and incoming orders, while delivery partners receive optimized dispatch assignments and route guidance. Administrators oversee the entire ecosystem through a unified dashboard.

1.3 Project Scope

The Online Food Ordering System shall include the following four portals within its scope:

- **Customer Portal:** Restaurant discovery, menu browsing, order placement, real-time delivery tracking, reviews, and loyalty rewards.
- **Restaurant Portal:** Menu management, order acceptance, kitchen display, revenue reporting, and promotion management.
- **Delivery Partner Portal:** Order assignment, route navigation, delivery confirmation, and earnings tracking.
- **Admin Portal:** Platform-wide user management, restaurant onboarding, dispute resolution, analytics, and configuration.

The following are explicitly OUT of scope for version 1.0:

- Dine-in table reservation or self-ordering kiosk integration
- Grocery or non-food item delivery
- Native mobile applications (Android/iOS) — web-responsive only
- Third-party loyalty scheme integrations (future roadmap)

1.4 Definitions, Acronyms & Abbreviations

OFOS	Online Food Ordering System
FRD	Functional Requirements Document
ETA	Estimated Time of Arrival
SKU	Stock Keeping Unit (menu item identifier)
POS	Point of Sale
OTP	One-Time Password
RBAC	Role-Based Access Control
SLA	Service Level Agreement
UPI	Unified Payments Interface
KOT	Kitchen Order Ticket
COD	Cash on Delivery
GSTIN	Goods and Services Tax Identification Number
EDD	Estimated Delivery Date/Time

1.5 Document Conventions

Requirements are uniquely identified using the format [MODULE]-[TYPE]-[NNN], where MODULE is a short prefix for the functional area, TYPE is FR (functional requirement), and NNN is a three-digit sequence. Priority levels are:

- **High:** Must be implemented in Version 1.0. Blocking for launch.
- **Medium:** Should be implemented in Version 1.0. Non-blocking but important.
- **Low:** Nice-to-have; may be deferred to a future release.

2. System Stakeholders & User Roles

2.1 Stakeholder Overview

Customer	End-user who browses restaurants, places food orders, makes payments, and tracks delivery.
Guest User	Unauthenticated visitor who can browse menus and restaurants but must register to place an order.
Restaurant Owner	Business owner who manages their restaurant profile, menu, operating hours, and views orders.

Restaurant Staff	Kitchen or front-desk staff who accept/reject incoming orders and update order status.
Delivery Partner	Independent rider/driver who picks up orders from restaurants and delivers to customers.
Fleet Manager	Manages and monitors the delivery partner pool, assigns orders manually if needed.
Platform Admin	Anthropic-side administrator who manages the entire platform including onboarding and config.
Finance Manager	Manages payouts to restaurants and delivery partners, reconciles transactions.

2.2 Role-Based Access Summary

Feature Area	Customer	Restaurant	Delivery	Fleet Mgr	Admin
Browse & Search Menus	✓	✓	—	—	✓
Place Orders	✓	—	—	—	—
Manage Menu / Items	—	✓	—	—	✓
Accept / Reject Orders	—	✓	—	—	✓
View Assigned Deliveries	—	—	✓	✓	✓
Manage Delivery Partners	—	—	—	✓	✓
Platform Reports	—	—	—	—	✓
Manage Promotions	—	✓	—	—	✓

3. System Architecture Overview

3.1 High-Level Architecture

The Online Food Ordering System follows a three-tier architecture composed of a presentation layer (React.js web front-end), a business logic layer (RESTful microservices), and a data layer (relational + NoSQL databases). Key architectural components include:

- **API Gateway:** Single entry point for all client requests, handling routing, rate limiting, and JWT authentication.
- **Order Service:** Core service managing order lifecycle from cart creation to delivery confirmation.
- **Catalog Service:** Manages restaurant profiles and menu item data with search indexing.
- **Delivery Service:** Handles delivery partner assignment, real-time location tracking, and ETA calculation.
- **Notification Service:** Orchestrates push, SMS, and email notifications across all user roles.
- **Payment Service:** Integrates with payment gateways and manages transaction records.
- **Analytics Service:** Aggregates platform data into dashboards and scheduled reports.

3.2 Technology Stack

Front-End	React.js with TypeScript; Tailwind CSS; Google Maps JavaScript SDK
Back-End	Node.js (Express) microservices; Python (FastAPI) for ML-based ETA and recommendations
Databases	PostgreSQL (transactional data); Redis (sessions, cart, real-time tracking); Elasticsearch (search)
Message Queue	RabbitMQ for asynchronous order and notification events
Payments	Razorpay (India); Stripe (International); PayPal (optional)
Maps & Tracking	Google Maps Platform (Directions API, Distance Matrix API, Maps JavaScript API)
Notifications	Firebase Cloud Messaging (push); Twilio (SMS); SendGrid (email)
Cloud	AWS (EC2, RDS, S3, CloudFront, SQS)
Authentication	JWT with OAuth 2.0 (Google, Facebook sign-in); bcrypt password hashing

4. Functional Requirements – Customer Portal

4.1 Registration & Authentication

This module covers how customers create accounts, authenticate, and manage their credentials.

Req. ID	Requirement Description	Priority	Category
CUST-F R-001	The system shall allow customers to self-register using full name, email address, mobile number, and password.	High	Authentication
CUST-F R-002	The system shall verify the customer's mobile number via a 6-digit OTP sent by SMS within 60 seconds of registration.	High	Authentication
CUST-F R-003	The system shall support login via Google OAuth 2.0 and Facebook Login as social sign-in options.	Medium	Authentication
CUST-F R-004	The system shall allow customers to log in using email/mobile and password, or via social sign-in.	High	Authentication
CUST-F R-005	The system shall provide a 'Forgot Password' flow via a time-limited (15-minute) secure reset link sent to the registered email.	High	Authentication
CUST-F R-006	The system shall lock a customer account after 5 consecutive failed login attempts and notify the customer via email with an unlock link.	High	Security
CUST-F R-007	Authenticated sessions shall expire after 30 days of inactivity; customers shall be prompted to re-authenticate.	Medium	Security
CUST-F R-008	Customers shall be able to view and terminate active sessions from other devices from their account settings.	Medium	Security

4.2 Customer Profile Management

Req. ID	Requirement Description	Priority	Category
CUST-F R-009	Customers shall be able to update their display name, profile photo, email, and mobile number.	High	Profile
CUST-F R-010	Customers shall be able to save multiple delivery addresses with labels (Home, Work, Other) and designate one as default.	High	Profile
CUST-F R-011	The system shall support address entry via: (a) manual text input, (b) GPS auto-detect (browser geolocation), or (c) Google Places Autocomplete search.	High	Profile
CUST-F R-012	Customers shall be able to store up to 5 payment methods (cards, UPI IDs) for faster checkout.	Medium	Profile
CUST-F R-013	Customers shall be able to configure dietary preferences (Vegetarian, Vegan, Gluten-Free, Halal, Jain) to filter menu items by default.	Medium	Profile

Req. ID	Requirement Description	Priority	Category
CUST-F R-014	Customers shall be able to set communication preferences for email, SMS, and push notifications independently.	Medium	Profile

4.3 Restaurant Discovery & Search

Customers must be able to efficiently find restaurants and food items relevant to their location and preferences.

Req. ID	Requirement Description	Priority	Category
CUST-F R-015	The system shall display a personalized homepage feed of restaurants sorted by relevance (based on order history, ratings, and proximity) when a delivery address is set.	High	Discovery
CUST-F R-016	Customers shall be able to search for restaurants or dishes by keyword; the system shall return results within 1.5 seconds using Elasticsearch.	High	Search
CUST-F R-017	The system shall support filtering search results by: cuisine type, price range (₹, ₹₹, ₹₹₹), average rating (3+, 4+, 4.5+), dietary tags, and delivery time range.	High	Search
CUST-F R-018	The system shall support sorting search results by: Relevance (default), Rating (high to low), Delivery Time (fastest first), and Price (low to high, high to low).	High	Search
CUST-F R-019	The system shall only display restaurants that deliver to the customer's active delivery address based on defined service radius polygons per restaurant.	High	Discovery
CUST-F R-020	The system shall display each restaurant card with: name, primary cuisine, average rating, number of reviews, estimated delivery time, minimum order value, and delivery fee.	High	Discovery
CUST-F R-021	The system shall display a 'Currently Closed' badge on restaurants outside their operating hours, and still allow customers to browse the menu.	High	Discovery
CUST-F R-022	Customers shall be able to view restaurants on an interactive map view with clustered pins representing nearby restaurants.	Low	Discovery

Req. ID	Requirement Description	Priority	Category
CUST-F R-023	The system shall provide a 'Top Picks Near You', 'Trending This Week', and 'Order Again' section on the homepage for returning customers.	Medium	Discovery

4.4 Menu Browsing & Item Selection

Req. ID	Requirement Description	Priority	Category
CUST-F R-024	The restaurant menu page shall display items grouped into categories (e.g., Starters, Mains, Beverages, Desserts) defined by the restaurant.	High	Menu
CUST-F R-025	Each menu item shall display: name, description (up to 200 characters), price, food type icon (Veg/Non-Veg/Vegan), photo, and availability status.	High	Menu
CUST-F R-026	Customers shall be able to filter menu items by dietary tags (Veg/Non-Veg/Vegan/Gluten-Free) directly within the restaurant menu page.	High	Menu
CUST-F R-027	Customers shall be able to customise items with add-ons and modifier groups (e.g., 'Choose base: Thin Crust / Thick Crust', 'Extra toppings') defined by the restaurant.	High	Menu
CUST-F R-028	The system shall enforce modifier group rules: minimum and maximum selections, mutually exclusive choices, and required vs. optional groups.	High	Menu
CUST-F R-029	Customers shall be able to add special instructions per item (free text, max 150 characters).	Medium	Menu
CUST-F R-030	The system shall display a prominent 'Bestseller' badge on items flagged as such by the restaurant.	Medium	Menu
CUST-F R-031	The system shall display an 'Out of Stock' overlay on items the restaurant has marked as unavailable, preventing them from being added to cart.	High	Menu
CUST-F R-032	The system shall allow customers to search for specific dishes within a restaurant's menu using a local search bar.	Medium	Menu

4.5 Shopping Cart

Req. ID	Requirement Description	Priority	Category
CUST-F R-033	The system shall maintain a persistent cart per customer per restaurant, surviving browser refresh and session changes (stored server-side via Redis).	High	Cart
CUST-F R-034	Customers shall be able to view a cart summary panel (slide-in drawer) without navigating away from the menu page.	High	Cart
CUST-F R-035	Customers shall be able to update item quantity or remove items from the cart at any point before checkout.	High	Cart
CUST-F R-036	The system shall warn customers when they attempt to add items from a different restaurant to an existing cart, offering the option to clear the current cart or cancel.	High	Cart
CUST-F R-037	The cart shall display a running subtotal, applicable taxes, delivery fee, any active discounts, and the final total, all updated in real time as items are added or removed.	High	Cart
CUST-F R-038	The system shall enforce a minimum order value per restaurant and display a progress bar showing how far the customer is from reaching the minimum.	High	Cart
CUST-F R-039	Customers shall be able to save the current cart as a 'Reorder Template' for future use.	Low	Cart

4.6 Checkout & Order Placement

Req. ID	Requirement Description	Priority	Category
CUST-F R-040	The checkout page shall present a final order summary, selected delivery address, delivery time estimate, payment method selector, and coupon/promo code input.	High	Checkout
CUST-F R-041	Customers shall be able to select from their saved addresses or add a new address during checkout.	High	Checkout
CUST-F R-042	Customers shall be able to choose between 'Deliver Now' (ASAP) and 'Schedule Delivery' (select date and time slot up to 48 hours in advance).	High	Checkout
CUST-F R-043	The system shall accept the following payment methods: Credit/Debit Card, UPI (via QR or UPI	High	Payment

Req. ID	Requirement Description	Priority	Category
	ID), Net Banking, Digital Wallets (Paytm, PhonePe), and Cash on Delivery (COD).		
CUST-F R-044	Customers shall be able to apply a coupon code during checkout; the system shall validate the coupon in real time and display the discount amount or a rejection reason.	High	Checkout
CUST-F R-045	The system shall display the delivery ETA prominently on the checkout page, calculated using the Google Maps Distance Matrix API.	High	Checkout
CUST-F R-046	Upon successful payment, the system shall generate a unique order ID and redirect the customer to an Order Confirmation page displaying order details, ETA, and a live tracking link.	High	Checkout
CUST-F R-047	The system shall send an order confirmation email and SMS to the customer within 30 seconds of order placement.	High	Notifications
CUST-F R-048	The checkout process shall complete (from 'Place Order' tap to confirmation screen) within 5 seconds under normal network conditions.	High	Performance
CUST-F R-049	In the event of payment failure, the system shall retain the cart contents and present a clear failure reason with a 'Retry Payment' option.	High	Checkout

4.7 Real-Time Order Tracking

This is a core differentiating feature — customers expect live, accurate visibility into their order status.

Req. ID	Requirement Description	Priority	Category
CUST-F R-050	The system shall display a live order tracking page with the following status stages: Order Placed → Restaurant Confirmed → Preparing → Ready for Pickup → Picked Up → Out for Delivery → Delivered.	High	Tracking
CUST-F R-051	The tracking page shall display a live map showing the delivery partner's current GPS location, the restaurant's pin, and the customer's delivery address.	High	Tracking
CUST-F R-052	The delivery partner's location on the tracking map shall update every 10 seconds via WebSocket push.	High	Tracking

Req. ID	Requirement Description	Priority	Category
CUST-F R-053	The system shall display an updated ETA on the tracking page, recalculated every 2 minutes based on current traffic and partner location.	High	Tracking
CUST-F R-054	Customers shall be able to call or send a pre-defined message to the delivery partner from the tracking page.	Medium	Tracking
CUST-F R-055	Customers shall receive a push notification and SMS when the order status changes to 'Picked Up' and 'Out for Delivery'.	High	Notifications
CUST-F R-056	The system shall send a 'Your order has arrived!' push notification when the delivery partner marks the order as Delivered.	High	Notifications
CUST-F R-057	The tracking page shall remain accessible for up to 24 hours post-delivery for reference.	Medium	Tracking

4.8 Order History & Reordering

Req. ID	Requirement Description	Priority	Category
CUST-F R-058	Customers shall be able to view a paginated order history listing all past orders with: order date, restaurant name, items ordered, total amount, and order status.	High	History
CUST-F R-059	Customers shall be able to view full details of any past order including itemized breakdown, delivery address, payment method, and invoice.	High	History
CUST-F R-060	Customers shall be able to download a PDF invoice for any completed order from the order details page.	Medium	History
CUST-F R-061	The system shall provide a one-tap 'Reorder' button on past orders that pre-populates the cart with the same items; the system shall flag any items no longer available.	High	History

4.9 Ratings, Reviews & Feedback

Req. ID	Requirement Description	Priority	Category
CUST-F R-062	Customers shall be prompted to rate their experience (1–5 stars for food quality, delivery	High	Reviews

Req. ID	Requirement Description	Priority	Category
	speed, and packaging) within 24 hours of a successful delivery.		
CUST-F R-063	Customers shall be able to write an optional text review (max 500 characters) and upload up to 3 food photos.	Medium	Reviews
CUST-F R-064	Reviews shall be submitted only for verified delivered orders linked to the customer's account (no anonymous reviews).	High	Reviews
CUST-F R-065	Customers shall be able to view all reviews for a restaurant, filtered by rating or sorted by most recent / most helpful.	Medium	Reviews
CUST-F R-066	Restaurants shall be able to post a public reply to customer reviews.	Medium	Reviews
CUST-F R-067	The system shall calculate and display the restaurant's overall rating as a weighted average of all verified reviews, updated in real time.	High	Reviews

4.10 Promotions, Coupons & Loyalty Rewards

Req. ID	Requirement Description	Priority	Category
CUST-F R-068	The system shall support multiple coupon types: percentage discount, flat amount discount, free delivery, BOGO (Buy One Get One), and minimum order amount discount.	High	Promotions
CUST-F R-069	The system shall enforce coupon rules: validity date range, minimum cart value, per-user usage limit, per-coupon total usage cap, and restaurant/category scope.	High	Promotions
CUST-F R-070	The system shall display all applicable active offers on the restaurant page and cart/checkout page automatically without requiring the customer to enter a code.	Medium	Promotions
CUST-F R-071	Customers shall earn loyalty points for every order (configurable rate per ₹/\$ spent) and be able to redeem points as a discount on future orders.	Medium	Loyalty
CUST-F R-072	Customers shall be able to view their current loyalty point balance and a transaction history of points earned and redeemed.	Medium	Loyalty
CUST-F R-073	The system shall support referral codes: existing customers receive loyalty points when a new	Low	Loyalty

Req. ID	Requirement Description	Priority	Category
	customer places their first order using the referral code.		

4.11 Cancellation & Refunds

Req. ID	Requirement Description	Priority	Category
CUST-F R-074	Customers shall be able to cancel an order without penalty only before the restaurant has accepted the order.	High	Cancellation
CUST-F R-075	After restaurant acceptance, order cancellation shall be subject to the platform's cancellation policy (partial or no refund based on preparation stage).	High	Cancellation
CUST-F R-076	Refunds for eligible cancellations shall be initiated within 24 hours of cancellation and credited to the original payment method within 5–7 business days.	High	Refunds
CUST-F R-077	Customers shall be able to raise a dispute for incorrect, missing, or poor-quality items from the order details page; the system shall generate a support ticket automatically.	High	Refunds
CUST-F R-078	The system shall display the refund status (Initiated, Processing, Credited) in the customer's order detail page.	Medium	Refunds

5. Functional Requirements – Restaurant Portal

5.1 Restaurant Onboarding & Profile

Req. ID	Requirement Description	Priority	Category
REST-F R-001	Restaurant owners shall register by providing: business name, GSTIN, FSSAI licence number, contact details, bank account details (for payouts), and at least one operational address.	High	Onboarding
REST-F R-002	The system shall place newly registered restaurants in a 'Pending Verification' state;	High	Onboarding

Req. ID	Requirement Description	Priority	Category
	admin approval is required before the restaurant goes live.		
REST-F R-003	Restaurant owners shall be able to edit their public profile: logo, cover photo, description (max 300 characters), cuisine tags, and contact number.	High	Profile
REST-F R-004	Restaurant owners shall be able to configure operating hours per day of week, with the option to set different hours for different days or mark a day as closed.	High	Profile
REST-F R-005	Restaurant owners shall be able to define a custom delivery radius (polygon or radius in km) per branch.	High	Profile
REST-F R-006	Restaurant owners shall be able to temporarily pause order acceptance (e.g., during a rush) and set an auto-resume time.	High	Operations
REST-F R-007	The system shall allow restaurant owners to manage multiple branches under a single account with separate menus and operating hours per branch.	Medium	Profile

5.2 Menu Management

Req. ID	Requirement Description	Priority	Category
REST-F R-008	Restaurant owners shall be able to create, edit, duplicate, and delete menu categories and menu items from a self-service dashboard.	High	Menu Mgmt
REST-F R-009	Each menu item shall support the following fields: name, description, price, food type (Veg/Non-Veg/Vegan), dietary tags, photo upload, and availability toggle.	High	Menu Mgmt
REST-F R-010	Restaurant owners shall be able to define modifier groups for items (e.g., 'Size': Small/Medium/Large) with optional extra charges per modifier.	High	Menu Mgmt
REST-F R-011	Restaurant owners shall be able to bulk-import menu items via a CSV upload template.	Medium	Menu Mgmt
REST-F R-012	Restaurant owners shall be able to mark individual items or entire categories as 'Unavailable' instantly, and the change shall propagate to customer-facing menus within 30 seconds.	High	Menu Mgmt

Req. ID	Requirement Description	Priority	Category
REST-F R-013	Restaurant owners shall be able to configure time-based item availability (e.g., Breakfast items available 6 AM – 11 AM only).	Medium	Menu Mgmt
REST-F R-014	Restaurant owners shall be able to set a 'Bestseller' badge on up to 5 items per category.	Low	Menu Mgmt
REST-F R-015	The system shall maintain a version history of menu changes with the ability to revert to a previous version.	Low	Menu Mgmt

5.3 Order Management

Req. ID	Requirement Description	Priority	Category
REST-F R-016	The restaurant dashboard shall display incoming orders in real time with an audible and visual alert for each new order.	High	Orders
REST-F R-017	Incoming orders shall display: order ID, customer name (first name + last initial), item list with customizations, special instructions, order type (ASAP/Scheduled), and payment status.	High	Orders
REST-F R-018	Restaurant staff shall be able to Accept or Reject an incoming order within 5 minutes; the system shall auto-reject and notify the customer if no action is taken.	High	Orders
REST-F R-019	Upon accepting an order, the system shall print (or display on screen) a Kitchen Order Ticket (KOT) with item details and a QR code linking to the order.	Medium	Orders
REST-F R-020	Restaurant staff shall be able to update the order status to 'Preparing', 'Ready for Pickup', and flag when the order is handed over to the delivery partner.	High	Orders
REST-F R-021	Restaurant staff shall be able to flag individual items as unavailable mid-order and notify the customer, offering a refund for the missing item.	High	Orders
REST-F R-022	The system shall display a daily order queue view filtered by status (Pending, In Progress, Ready, Completed, Cancelled).	High	Orders
REST-F R-023	Restaurant owners shall be able to view historical orders with search and filter capabilities (by date, status, order ID, and payment method).	High	Orders

5.4 Promotions & Offers

Req. ID	Requirement Description	Priority	Category
REST-F R-024	Restaurant owners shall be able to create restaurant-specific promotions (percentage off, flat discount, free delivery) with configurable validity dates, minimum order values, and usage limits.	High	Promotions
REST-F R-025	Restaurants shall be able to feature a 'Deal of the Day' or limited-time offer that is prominently displayed on their restaurant card and menu page.	Medium	Promotions
REST-F R-026	Restaurant owners shall be able to view coupon performance metrics: redemption count, total discount granted, and average order value with coupon.	Medium	Promotions

5.5 Analytics & Reporting

Req. ID	Requirement Description	Priority	Category
REST-F R-027	The restaurant dashboard shall display real-time KPIs: orders today, revenue today, average order value, and current order acceptance rate.	High	Analytics
REST-F R-028	Restaurant owners shall be able to view revenue reports by: day, week, and month, with trend graphs showing comparison to the previous period.	High	Analytics
REST-F R-029	The system shall provide a 'Top Selling Items' report showing units sold, revenue contribution, and average rating per item.	Medium	Analytics
REST-F R-030	The system shall provide a customer demographics report showing order frequency, new vs. returning customer split, and peak ordering hours (heatmap).	Medium	Analytics
REST-F R-031	Restaurant owners shall be able to export all reports as PDF or CSV files.	Medium	Analytics
REST-F R-032	Restaurant owners shall be able to view their pending payout amount, payout schedule, and transaction history with commission deductions itemized.	High	Finance

6. Functional Requirements – Delivery Partner Portal

6.1 Delivery Partner Onboarding

Req. ID	Requirement Description	Priority	Category
DEL-FR-001	Delivery partners shall register by providing: full name, mobile number, vehicle type (Bicycle/Motorcycle/Car), vehicle registration number, and a government-issued ID.	High	Onboarding
DEL-FR-002	The system shall require admin verification of submitted documents before activating a delivery partner account.	High	Onboarding
DEL-FR-003	Delivery partners shall be able to set their availability status to Online (accepting orders) or Offline (not available) at any time.	High	Operations

6.2 Order Assignment & Dispatch

Req. ID	Requirement Description	Priority	Category
DEL-FR-004	The system shall automatically assign incoming ready-for-pickup orders to the nearest available online delivery partner using a proximity-based algorithm.	High	Dispatch
DEL-FR-005	Delivery partners shall receive a push notification and in-app alert for each new order assignment containing: restaurant name, pickup address, customer area, and estimated earnings.	High	Dispatch
DEL-FR-006	Delivery partners shall be able to Accept or Decline an assigned order within 60 seconds; declined orders shall be automatically reassigned to the next available partner.	High	Dispatch
DEL-FR-007	After 2 consecutive declines by different partners, the order shall be escalated to the Fleet Manager for manual assignment.	Medium	Dispatch
DEL-FR-008	Fleet managers shall be able to manually assign or reassign orders to specific delivery partners from the fleet management dashboard.	High	Dispatch

6.3 Navigation & Delivery Execution

Req. ID	Requirement Description	Priority	Category
DEL-FR-009	After accepting an order, the delivery partner app shall display a turn-by-turn navigation route to the restaurant's pickup address using Google Maps Directions API.	High	Navigation
DEL-FR-010	After confirming order pickup, the system shall display a navigation route to the customer's delivery address.	High	Navigation
DEL-FR-011	Delivery partners shall be able to confirm delivery by tapping a 'Delivered' button; for high-value orders, the system shall require a delivery confirmation photo or customer OTP.	High	Delivery
DEL-FR-012	The delivery partner's GPS location shall be transmitted to the platform every 10 seconds when they have an active order.	High	Tracking
DEL-FR-013	Delivery partners shall be able to report issues (e.g., customer unavailable, wrong address, restaurant delay) from the order screen; each report shall auto-notify the customer and support team.	High	Operations

6.4 Earnings & Performance

Req. ID	Requirement Description	Priority	Category
DEL-FR-014	The delivery partner app shall display a real-time earnings dashboard showing: earnings today, this week, this month, number of deliveries completed, and tips received.	High	Earnings
DEL-FR-015	Delivery partners shall be able to view an itemized earning history per delivery (base pay + distance bonus + tips).	Medium	Earnings
DEL-FR-016	The system shall calculate and display a delivery partner's performance rating (based on on-time delivery rate, customer ratings, and acceptance rate).	Medium	Performance
DEL-FR-017	The system shall process weekly payouts to verified delivery partner bank accounts; partners shall receive a payout confirmation notification.	High	Earnings

7. Functional Requirements – Admin Portal

7.1 User & Restaurant Management

Req. ID	Requirement Description	Priority	Category
ADMIN-FR-001	Admins shall be able to view, search, and filter all user accounts (customers, restaurant owners, delivery partners) and view detailed profile information.	High	User Mgmt
ADMIN-FR-002	Admins shall be able to approve, reject, suspend, or permanently deactivate any user account with a mandatory reason field.	High	User Mgmt
ADMIN-FR-003	Admins shall be able to review and approve restaurant onboarding applications with the ability to request additional documentation.	High	Onboarding
ADMIN-FR-004	Admins shall be able to manually override restaurant menu item availability, operating hours, or temporarily delist a restaurant.	High	Restaurant Mgmt
ADMIN-FR-005	Admins shall be able to create, edit, and deactivate platform-wide coupon codes and promotional campaigns.	High	Promotions

7.2 Order Monitoring & Dispute Management

Req. ID	Requirement Description	Priority	Category
ADMIN-FR-006	Admins shall have a real-time order monitoring dashboard displaying all active orders across the platform with current status, restaurant, delivery partner, and elapsed time.	High	Monitoring
ADMIN-FR-007	Admins shall receive alerts for orders that have been in the same status for longer than a configurable threshold (e.g., 'Preparing' for > 45 minutes).	High	Monitoring
ADMIN-FR-008	Admins shall be able to view, respond to, and resolve customer dispute tickets, and trigger partial or full refunds with a reason code.	High	Disputes
ADMIN-FR-009	The system shall log all admin actions (with user ID, timestamp, and action details) in an immutable audit trail.	High	Compliance

7.3 Platform Analytics & Reporting

Req. ID	Requirement Description	Priority	Category
ADMIN-FR-010	The admin dashboard shall display platform-wide KPIs: gross merchandise value (GMV), total orders, active customers, new restaurants, average delivery time, and cancellation rate — all filterable by date range.	High	Analytics
ADMIN-FR-011	Admins shall be able to generate financial reconciliation reports showing gross sales, platform commission, restaurant payouts, and delivery partner payouts by period.	High	Finance
ADMIN-FR-012	The system shall provide a geographic heatmap of order density across service areas, updatable by date range.	Medium	Analytics
ADMIN-FR-013	Admins shall be able to schedule automated report delivery to configured email recipients (daily, weekly, monthly).	Medium	Reporting

7.4 Platform Configuration

Req. ID	Requirement Description	Priority	Category
ADMIN-FR-014	Admins shall be able to configure global platform settings: default delivery radius, platform commission percentage per restaurant tier, and minimum delivery fee.	High	Config
ADMIN-FR-015	Admins shall be able to manage notification templates (email, SMS, push) for all system events from a WYSIWYG template editor.	Medium	Config
ADMIN-FR-016	Admins shall be able to manage the list of supported cuisine types, dietary tags, and city/service-area boundaries.	Medium	Config
ADMIN-FR-017	Admins shall be able to configure the loyalty points earning rate (points per ₹ spent) and redemption rate (₹ value per point).	Medium	Config

8. Functional Requirements – Cross-Cutting Concerns

8.1 Notifications Architecture

Event	Customer	Restaurant	Delivery Partner
Order Placed	Email + SMS + Push	Push + Dashboard Alert	—
Order Accepted	Push + SMS	—	—
Order Rejected	Push + SMS	—	—
Order Ready for Pickup	—	—	Push + SMS
Order Picked Up	Push + SMS	—	—
Out for Delivery	Push + SMS	—	—
Order Delivered	Push + SMS	—	Push
Refund Initiated	Email + SMS	—	—
Payout Processed	—	Email	Email + SMS
New Review Posted	—	Push + Email	—

8.2 Payment & Financial Processing

Req. ID	Requirement Description	Priority	Category
PAY-FR-001	The system shall integrate with Razorpay and Stripe as primary payment gateways, with automatic failover between gateways if one is unavailable.	High	Payments
PAY-FR-002	The system shall support PCI-DSS compliant card tokenization; raw card details shall never be stored on platform servers.	High	Security
PAY-FR-003	The system shall apply platform commission (configurable per restaurant tier) to each order and compute the net amount payable to the restaurant.	High	Finance
PAY-FR-004	The system shall batch weekly payouts to restaurants and delivery partners, deducting applicable commissions and adjusting for any outstanding refunds.	High	Finance
PAY-FR-005	All financial transactions shall be logged immutably with: transaction ID, amount, currency, gateway response code, and timestamp.	High	Compliance

8.3 Search & Recommendation Engine

Req. ID	Requirement Description	Priority	Category
SRCH-FR-001	The search engine shall support typo-tolerance (fuzzy matching within 2 character edits) for restaurant and dish searches.	High	Search
SRCH-FR-002	Search results shall be ranked using a composite score of: text relevance, proximity, rating, order volume, and whether the restaurant is currently open.	High	Search
SRCH-FR-003	The system shall provide predictive search suggestions as the customer types, based on popular searches and the customer's order history.	Medium	Search
SRCH-FR-004	The recommendation engine shall suggest restaurants on the homepage based on collaborative filtering (customers with similar order histories).	Low	Recommendations

9. Detailed Use Cases

UC-01: Customer Places a New Order

Use Case ID	UC-01
Actor	Registered Customer
Goal	Customer successfully places a food order and receives confirmation.
Pre-conditions	Customer is logged in. Delivery address is set. Restaurant is open and within service radius.
Main Flow	1. Customer searches for a restaurant or dish by keyword or browses the homepage feed. 2. Customer selects a restaurant and views the menu. 3. Customer adds desired items to cart, configuring customisations and special instructions as needed. 4. Customer reviews the cart, confirms the order summary and delivery address. 5. Customer applies a promo code (optional); discount is applied automatically. 6. Customer selects a payment method and taps 'Place Order'. 7. Payment is processed by the gateway; the system receives a success callback.

	8. System creates the order record, assigns an order ID, and notifies the restaurant. 9. Customer is redirected to the Order Confirmation page with ETA and a live tracking link. 10. System sends confirmation email and SMS to the customer.
Alternate Flow A	Step 7 – Payment fails: system displays a failure message with reason, cart is retained, customer can retry with same or different payment method.
Alternate Flow B	Step 3 – Item is out of stock: system shows 'Unavailable' overlay; customer cannot add it to cart.
Alternate Flow C	Step 3 – Cart already has items from a different restaurant: system prompts 'Replace cart?' dialog.
Post-conditions	Order is created with status 'Pending'. Restaurant receives a new order notification. Customer receives confirmation.

UC-02: Restaurant Accepts & Fulfills an Order

Use Case ID	UC-02
Actor	Restaurant Staff
Goal	Restaurant processes an incoming order through to delivery handoff.
Pre-conditions	Restaurant is online. A customer has placed an order.
Main Flow	1. New order alert sounds on the restaurant dashboard; KOT is displayed. 2. Staff reviews order details and taps 'Accept Order' within 5 minutes. 3. System updates order status to 'Accepted' and notifies the customer. 4. Kitchen prepares the order; staff updates status to 'Preparing'. 5. When food is ready, staff taps 'Ready for Pickup'. 6. System notifies the assigned delivery partner that the order is ready. 7. Delivery partner arrives; staff confirms handoff and taps 'Handed Over'. 8. Order status updates to 'Picked Up'.
Alternate Flow	Step 2 – Staff rejects the order: system prompts for a rejection reason, customer is notified and refunded automatically.

Post-conditions	Order is in 'Picked Up' state. Delivery partner has started navigation to the customer.
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UC-03: Delivery Partner Completes a Delivery

Use Case ID	UC-03
Actor	Delivery Partner
Goal	Delivery partner successfully delivers the order to the customer.
Pre-conditions	Delivery partner is online. Order has been assigned and accepted by the partner.
Main Flow	1. Delivery partner receives order assignment with restaurant address and map navigation. 2. Partner navigates to restaurant, picks up the order, and taps 'Order Picked Up'. 3. System displays navigation route to customer's delivery address. 4. Partner delivers the order to the customer. 5. Partner taps 'Delivered'; system prompts for customer OTP (for high-value orders) or delivery photo. 6. Partner enters OTP or uploads photo; system validates and marks order as Delivered. 7. Customer receives 'Order Delivered' notification. 8. Earnings for the delivery are credited to partner's account balance.
Alternate Flow	Step 5 – Customer is unavailable: partner selects 'Customer Not Available'; system triggers a 3-attempt protocol and escalates to support if unresolved.
Post-conditions	Order is marked Delivered. Customer is prompted for review. Partner's earnings are updated.

UC-04: Customer Raises a Refund Dispute

Use Case ID	UC-04
Actor	Customer, Admin
Goal	Customer receives a refund for a missing or incorrect item.
Pre-conditions	Order is in Delivered status within the last 24 hours.

Main Flow	1. Customer opens order history and selects the delivered order. 2. Customer taps 'Report an Issue' and selects issue type (Missing Item / Wrong Item / Poor Quality). 3. Customer selects the affected items, writes a brief description, and optionally uploads a photo. 4. System creates a support ticket and auto-credits 50% of the item value as a goodwill credit if the issue type is 'Missing Item'. 5. Ticket is assigned to an admin for review. 6. Admin reviews the ticket, approves a full or partial refund, and closes the ticket. 7. Customer receives a notification that the refund has been initiated. 8. Refund is credited to the original payment method within 5–7 business days.
Post-conditions	Support ticket is resolved. Refund is processed. Customer's experience is logged for quality tracking.

10. Business Rules

10.1 Ordering Rules

- **BR-ORD-01:** A customer may not have more than 3 active (undelivered) orders simultaneously.
- **BR-ORD-02:** Scheduled orders must be placed at least 1 hour in advance of the desired delivery time.
- **BR-ORD-03:** If a restaurant does not accept an order within 5 minutes, the system auto-rejects it, notifies the customer, and initiates a full refund.
- **BR-ORD-04:** An order once in 'Preparing' status cannot be cancelled by the customer; only admin can cancel with a partial refund determination.

10.2 Delivery Rules

- **BR-DEL-01:** A delivery partner may not have more than 2 concurrent active deliveries.
- **BR-DEL-02:** Delivery partners must maintain an acceptance rate of $\geq 70\%$ to retain active status; below this, the system sends a warning, and below 50%, the account is reviewed.
- **BR-DEL-03:** Delivery proof (OTP or photo) is mandatory for orders with a cart value exceeding ₹1,000 / \$15.

10.3 Payment & Refund Rules

- **BR-PAY-01:** Cash on Delivery (COD) is only available for orders below ₹2,000 / \$25.
- **BR-PAY-02:** Platform commission ranges from 15% to 30% of order subtotal, based on the restaurant's subscription tier.
- **BR-PAY-03:** Refunds are only eligible for disputes raised within 24 hours of the delivery timestamp.
- **BR-PAY-04:** Loyalty points cannot be redeemed simultaneously with a percentage-off coupon; only one discount type may apply per order.

10.4 Review Rules

- **BR-REV-01:** Only customers with a verified delivered order linked to their account may submit a review for that restaurant.
- **BR-REV-02:** Reviews may be edited once within 48 hours of submission; after 48 hours, reviews are locked.
- **BR-REV-03:** Reviews containing profanity or links shall be automatically flagged for admin moderation before being published.

11. Assumptions & Constraints

11.1 Assumptions

1. All restaurants have a stable internet connection sufficient to receive and manage orders in real time.
2. Delivery partners carry a smartphone with GPS enabled and the mobile web app loaded at all times while on duty.
3. Customers have a valid email address and mobile phone number for verification and notifications.
4. Payment gateway uptime is governed by the respective third-party SLA (Razorpay: 99.9%, Stripe: 99.99%).
5. Google Maps Platform API will maintain availability and accuracy of geolocation and routing data.
6. Restaurants are responsible for maintaining up-to-date menu information and item availability.

11.2 Constraints

- **Technical:** The system must be designed as a responsive web application; native mobile apps are out of scope for v1.0.
- **Regulatory:** The platform must comply with applicable data protection regulations (DPDP Act 2023 in India, GDPR for international expansion).
- **Financial:** PCI-DSS compliance is mandatory; card data must never be stored on the platform's own servers.

- **Geographic:** Version 1.0 is limited to restaurant delivery within India; international expansion is a future roadmap item.
- **Language:** Version 1.0 supports English and Hindi; additional regional languages may be added based on market demand.
- **Browser:** The web application must support the latest two versions of Chrome, Firefox, Safari, and Edge.

12. Glossary

Active Order	An order in any status between 'Order Placed' and 'Out for Delivery' (inclusive).
Cart	A temporary collection of items a customer intends to purchase, associated with one restaurant.
Commission	Platform's percentage cut of each order's subtotal, deducted before restaurant payout.
ETA	Estimated Time of Arrival — the predicted time for the order to reach the customer.
GMV	Gross Merchandise Value — total order value transacted through the platform before deductions.
KOT	Kitchen Order Ticket — a printed or displayed summary of an order's items for kitchen staff.
Modifier Group	A configurable set of options customers can choose from when customising a menu item.
Service Radius	The geographic area (defined by polygon or radius) within which a restaurant accepts deliveries.
Soft Delete	Deactivating a record without permanently removing it from the database, preserving historical data.
Tier	A restaurant subscription level that determines the platform commission rate and feature access.
WebSocket	A full-duplex communication protocol used for real-time data push (order status, GPS location).
Verified Delivery	A delivery confirmed by customer OTP or partner-uploaded photo.

13. Revision History

Version	Date	Author	Description
1.0	May 2026	Product & BA Team	Initial version of the Functional Requirements Document.

— End of Functional Requirements Document —